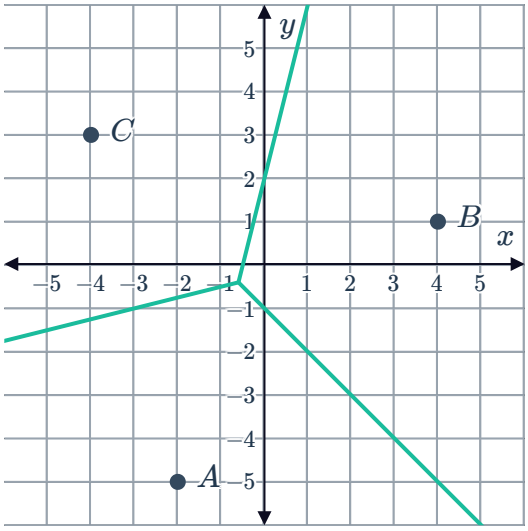
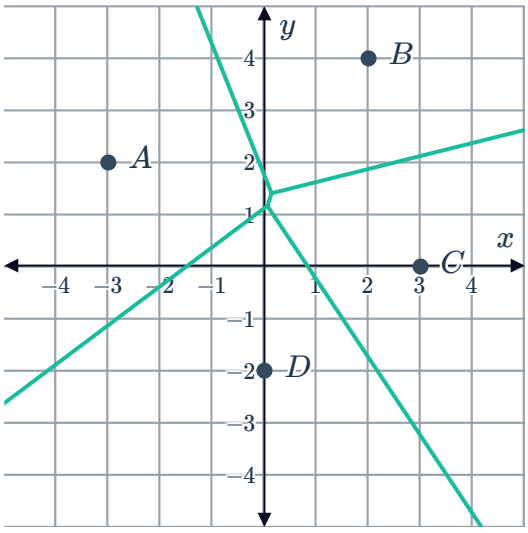
# Voronoi diagrams

**1** Consider the following Voronoi diagram for sites $\mathrm{A}$, $\mathrm{B}$ and $\mathrm{C}$:



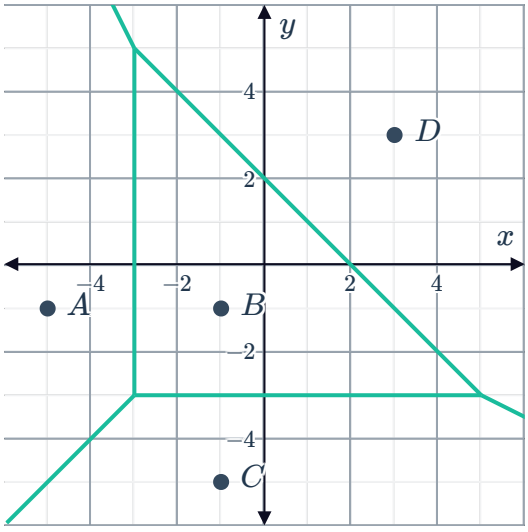

- **a** What can be said about all the points in cell $\mathrm{A}$?
- **b** What can be said about all the points along the edge between cell $\mathrm{A}$ and cell $\mathrm{B}$?
- **c** What can be said about the vertex?

**2** Consider the following Voronoi diagram:




- **a** Which site is the point $(-3, 5)$ closest to? Explain your answer.
- **b** Verify your answer to part (a) by finding the distance between point $(-3, 5)$ and each of the sites.
- **c** Identify the site which is closest to the following points:
  - **i** $(1, 4)$
  - **ii** $(-3, 0)$
  - **iii** $(3, -3)$
  - **iv** $(1, 2)$

**3** Consider the following Voronoi diagram:

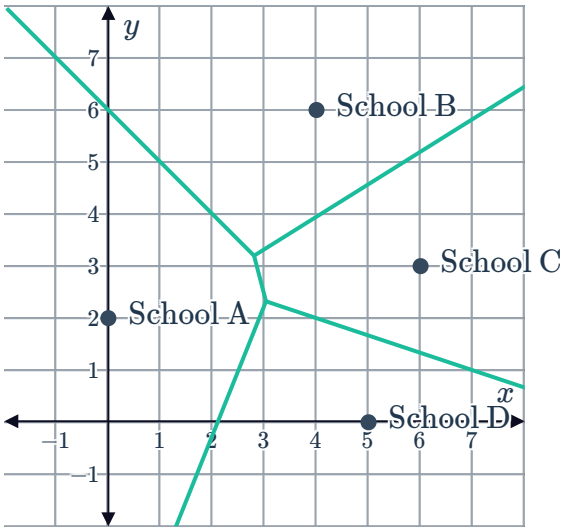



- **a** Show that $(-3, 0)$ is equidistant to sites $\mathrm{A}$ and $\mathrm{B}$.
- **b** Show that $(3, -1)$ is equidistant to sites $\mathrm{B}$ and $\mathrm{D}$.
- **c** Show that $(5, -3)$ is equidistant to sites $\mathrm{B}$, $\mathrm{C}$ and $\mathrm{D}$.
- **d** Show that $(-3, -3)$ is equidistant to sites $\mathrm{A}$, $\mathrm{B}$ and $\mathrm{C}$.
- **e** Does there exist a point that is equidistant to all four sites? Explain your answer.

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## Applications

**4** The following Voronoi diagram shows the locations of four schools in a city. Families are encouraged to send their children to their local (nearest) school. Each unit on the graph represents $1$ km.




**a** Yosef's house is located at $(3, 0)$. Which school should he attend?

**b** Ashleigh lives at $(1, 4)$.

**i** Which school should she attend?

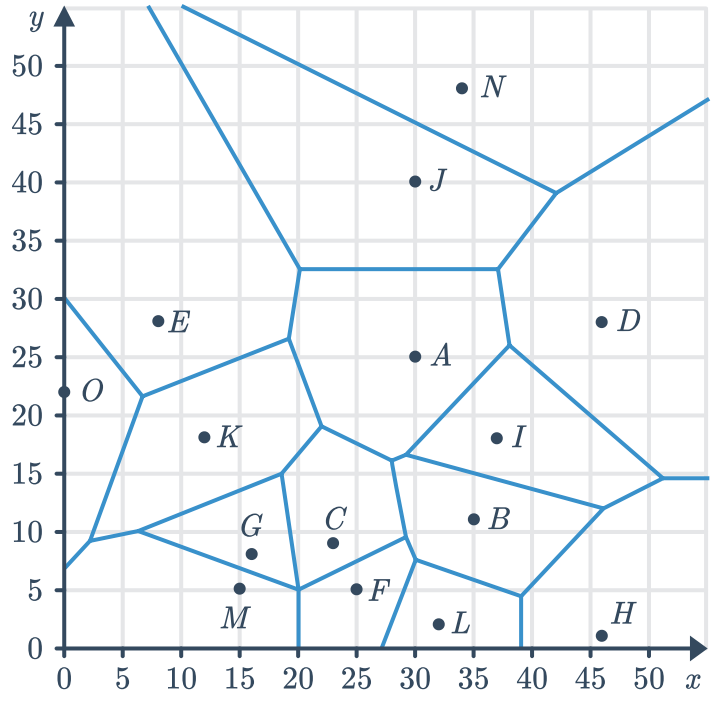
**ii** Ashleigh really wants to attend School C because that is where most of her friends are going. How much further will she have to travel to attend school C, compared to her local school? Round your answer to two decimal places.

**c** Han lives at $(4, 2)$.

**i** Which school should he attend?

**ii** Han and his family move $3$ km West. Which school should he attend now?

**5** The following Voronoi diagram shows a map of towns, where each unit is a kilometre. We need to find a location for a waste dump, however none of the towns want to be near it.




Suggest a location for the waste dump somewhere on this map. Explain your choice.

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